

Labwork 2: GPU Information Report

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1 Introduction

This report describes a Python program used to obtain basic information about available NVIDIA GPUs using the Numba CUDA interface. The program identifies the active GPU and then iterates through all available GPUs to collect hardware information.

2 Method

The program uses the following Numba CUDA functionality:

- `cuda.get_current_device()` to identify the currently active GPU.
- `cuda.gpus` to iterate over all available GPUs.
- `MULTIPROCESSOR_COUNT` to obtain the number of streaming multiprocessors (SMs).
- `compute_capability` to obtain the GPU compute capability.
- `get_memory_info()` to determine the available and total GPU memory.

3 Output

For each detected GPU, the program reports the following information:

Parameter	Description
GPU ID	Identifier of the GPU device
Name	Model name of the GPU
SM Count	Number of streaming multiprocessors
Compute Capability	CUDA compute capability version
Estimated CUDA Cores	Estimated number of CUDA cores
Total Memory	Total GPU memory in GB and bytes

Table 1: GPU information collected by the program

```
Device Name: Tesla T4
GPU ID: 0
Name: Tesla T4
Multiprocessor Count (SMs): 40
Compute capability: ComputeCapability(major=7, minor=5)
Estimated CUDA Core Count: 2560
Total Memory Size: 14.56 GB (15637086208 bytes)
```

Figure 1: Output of the GPU information program